ModelIt!

Student Activity Pack

Why Do Invasive Species Win?

Modeling Ecological Competition, Niche Displacement, and Biodiversity Collapse

Grade 11L2 | NGSS HS-LS2-2, HS-LS2-6, HS-LS4-5 | Cause and Effect / Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think makes a species 'invasive' — is every non-native species invasive, or only some? What's the difference?**

2. **Why might a species from another continent outcompete organisms that have lived in an ecosystem for thousands of years?**

3. **Draw a diagram showing what you think happens to a food web when a new aggressive predator or competitor is added.**

4. **Can you think of any invasive species in your area — what are they, and what effects have you noticed?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Invasive Species Introduction Rate
- Invasive Population Growth Rate
- Native Species Competitive Fitness
- Resource Availability
- Native Biodiversity Index
- Ecosystem Function Integrity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Early Detection

100 invasive individuals, diverse native community intact

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Established Invasion

Invasive population above establishment threshold, spreading

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Managed Ecosystem

Biocontrol agents introduced plus active removal of invasive biomass

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Disease and Parasite Introduction — Invasive species often carry pathogens and parasites to which native species have no immunity — introduced diseases have caused some of the most devastating ecological collapses, including chytrid fungus driving global amphibian extinctions and avian malaria devastating Hawaiian birds

Habitat Modification Feedback — Many invasive species actively alter their environment in ways that favor their continued spread — invasive grasses increase fire frequency and intensity, which kills fire-intolerant native trees and creates more grassland for the invader, a classic invasion-fire feedback loop

Genetic Hybridization — Some invasive species interbreed with closely related native species, producing hybrids that dilute the native gene pool and can create 'super-invaders' with combined traits of both parent species — genetic swamping is an underrecognized extinction mechanism

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What gives invasive species their competitive edge — and why do native ecosystems, shaped by millions of years of evolution, crumble so quickly when invaders arrive? Use evidence from your simulation to support your answers.

1. **Why does your model show that the establishment threshold is the most critical point in an invasion — and what does this mean for where we invest management resources?**

2. **How do positive feedback loops in your model explain why invasions accelerate rather than stabilize once a certain population density is reached?**

3. **What does the model reveal about why island ecosystems and isolated habitats are disproportionately devastated by invasive species?**

4. **How does the loss of a single keystone species to invasive competition cascade through an entire food web in your model?**

5. **What does your model suggest about the ethical complexity of invasive species management — killing one species to save others, introducing biocontrol agents that might themselves become invasive?**

8 STEM Challenge: Design an Invasive Species Early Detection and Rapid Response System

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YOUR ENGINEERING MISSION

Design a comprehensive monitoring and rapid response system for a vulnerable ecosystem that can detect invasive species introductions within 30 days and mobilize effective containment before populations cross the establishment threshold.

THE SCENARIO

A nationally significant nature preserve containing 15 endangered endemic species has experienced three invasive species introductions in the past decade, each discovered too late for eradication. The preserve management board needs a detection and response system that identifies new invasions within weeks — not years — and triggers rapid containment. Your team must design the monitoring technology, response protocols, and decision framework.

Engineering Design Thinking

1. HOW WOULD YOU PRIORITIZE MONITORING EFFORT ACROSS THE PRESERVE — WHICH AREAS AND INTRODUCTION PATHWAYS POSE THE HIGHEST INVASION RISK?

2. WHAT SAMPLING FREQUENCY AND SPATIAL COVERAGE WOULD YOUR MONITORING NETWORK REQUIRE TO DETECT A NEW SPECIES WITHIN 30 DAYS OF INTRODUCTION?

3. HOW WOULD YOU DECIDE BETWEEN ERADICATION, CONTAINMENT, AND LONG-TERM MANAGEMENT FOR DIFFERENT INVASION SCENARIOS?

4. WHAT PARTNERSHIPS WITH AGENCIES, UNIVERSITIES, AND COMMUNITY VOLUNTEERS WOULD YOUR SYSTEM REQUIRE TO SUSTAIN MONITORING AND RESPONSE CAPACITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Conservation Biologists study biodiversity patterns and develop strategies to protect endangered species and ecosystems from threats including invasive species, habitat loss, and climate change. They work for organizations like the U.S. Fish and Wildlife Service, The Nature Conservancy, and national parks, earning $55,000–$110,000/year. Invasion Ecologists specializing in invasive species detection, management, and ecosystem restoration earn $60,000–$125,000/year at research universities and environmental agencies.